

SUMMARY

Highly motivated and detail-oriented Computer Engineering student from National University of Singapore seeking an internship opportunity in IoT Technologies, Robotics and Data Analytics. Thrives in collaborative team environments, demonstrating an unwavering commitment to both personal and collective success. Possesses a strong academic foundation and an ardent passion for exploring the realms of IoT, Robotics and practical applications.

EDUCATION

National University of Singapore

Bachelor's in Engineering

Computer Engineering

2025 - Present

Cumulative GPA: 3.7

Clubs and Societies: Raffles Hall Voices,

RH Green Comm, RH Tech Crew, RH

Board of Photography

Singapore Polytechnic

Diploma in Computer Engineering

Specialization in Smart City Technologies

2022 - 2025

GPA: 3.94

Clubs and Societies: Infocomm Club, SP

Primers

SKILLS

- **Programming Languages:** C, C++, Python, R, JavaScript (Node.js, React.js), PHP, HTML/CSS
- **Robotics & Embedded Systems:** Microcontrollers (PIC, Arduino Uno, Arduino Mega), Raspberry Pi
- **Networking & Cloud:** TCP/UDP Protocols, MQTT, AWS Cloud Services
- **Data Analysis:** RStudio, Radiant, Spotfire

ACHIEVEMENTS

- **Lee Hsien Loong Interactive Digital Media Smart Nation Award 2025 (Top 25):** Built an AI-based glass detection system using computer vision and AWS SNS.
- **CDL Young SDG Leaders (Top 10):** Partnered with Schneider Electric to design Scope 3 decarbonization and bioleaching strategies.
- **Director's Honour Roll, Singapore Polytechnic (2024):** Awarded to top 10% of the Cohort for outstanding academic performance

WORK EXPERIENCE

Internship at Singapore Polytechnic (5G & AIoT Centre)

Singapore Polytechnic | Mar 2024 - Aug 2024

- Developed an IoT Platform for sensor data collection using microservices architecture with **ReactJS frontend and NodeJS backend**
- Migrated the deployment to the **AWS Cloud**, while implementing **auto-scaling** features to improve scalability
- Automated the deployment and created a Continuous Integration/Continuous Deployment (**CI/CD**) Pipeline, simplifying future development
- Reduced **cost of maintenance and manpower** of the IoT Platform by more than **50%**

PROJECTS

- Tele-operated Rescue Robot featuring a distributed **dual-operator TCP** architecture across a Raspberry Pi and Arduino Mega, utilizing multi-processed **SLAM mapping**, bare-metal **interrupt-driven** hardware control for zero-latency navigation and **4-DOF Robot Arm** control.
- **Automatic Seeder** with Ultrasonic sensor and IR guided pathway using Arduino Uno
- Home automation system with **Voice Control** and **Remote Control** through websites using Raspberry Pi
- Created a cycling-initiative which utilizes a **rewards based** system for users, by unifying a **dynamic website, mobile application and microcontroller** to promote a cycling culture in Singapore.

LEADERSHIP AND VOLUNTEERING

Head of Choir | Raffles Hall, NUS | Mar 2026 - Present

- Direct timeline planning, organization-wide communications, and general administrative duties for Raffles Hall Choir.
- Guide the overall direction of the CCA, liaising with the Junior Common Room Committee (JCRC) for event execution while actively prioritizing member wellbeing and engagement.

Boys Brigade Share-A-Gift | Singapore Polytechnic | Nov 2023 - Dec 2023

- Facilitated door-to-door food donation collections, collaborating in a team to execute community outreach initiatives supporting the less fortunate.
- Engaged effectively with a diverse range of community residents to advocate for the charity initiative and encourage public participation.